Virtual pH Sensor for Hydroponics

Confidence-Aware GRU + CARD — Summary Report

Reproduction of Moniruzzaman et al. (AEJ 2026) + research extension | Kaggle DOI 10.34740/kaggle/ds/8223904

Project in one paragraph

We build a virtual pH sensor that predicts hydroponic water pH from cheaper sensors, so the fragile physical pH probe is used less and lasts longer. We reproduced the paper (R^2=0.98), found that score was inflated by temporal data leakage, built an honest leakage-free test, fixed the model so it generalises, and added confidence-based sensor switching, fault robustness, and a budget-targeting controller (CARD).

Model comparison — HONEST leakage-free test (ranked by R^2)

Model	R^2	MAE	RMSE	within ±0.05
ExtraTrees	0.676	0.0199	0.0253	95.1%
Persistence	0.650	0.0196	0.0263	93.8%
CARD (ours)	0.650	0.0196	0.0263	94.5%
Improved GRU	0.650	0.0196	0.0263	94.4%
XGBoost	0.550	0.0236	0.0298	90.9%
RandomForest	0.262	0.0309	0.0381	81.5%
LinearRegression	0.194	0.0326	0.0399	80.1%
LSTM	0.172	0.0355	0.0404	77.0%
GRU	0.030	0.0390	0.0437	72.3%

Reading: original GRU fails (R^2=0.03) on the honest test; our Improved GRU & CARD recover to the top group (0.65) at half the parameters; ExtraTrees is the best point-accuracy model. Note a trivial 'persistence' baseline reaches ~0.65 too — so raw accuracy is not our contribution.

Key measured results

Physical-probe usage cut	75.64766839378238%
Fault: proposed vs traditional R^2	0.89 vs -6.9
Model footprint	10,733 params · ~0.9 ms · 6.8 KB ONNX

Honest interpretation & practical impact

Is the 94% good?

94.5% of predictions land within ±0.05 pH — measured honestly. But this is a within-tolerance rate, NOT accuracy, and a trivial 'repeat-last-value' predictor also reaches 93.8% because pH moves slowly. So the 94% is not our contribution. Our value is knowing when it's unsure, cutting probe usage 76%, and surviving sensor faults — which a trivial predictor cannot.

Limitations (stated openly)

- Single ~1-month hydroponic run — no cross-crop/season test yet.
- Persistence is near-ceiling on this smooth signal; we match not dominate it on raw accuracy.
- Sensor faults (Phase 8) are realistic but simulated, not recorded.
- One-step-ahead only; multi-step forecasting is the key next experiment.
- Probe lifespan is ESTIMATED (usage-proportional wear), not measured.
- CPU-only training (hardware-independent for these tiny models).

Cost & sensor-lifespan ESTIMATE (assumptions stated)

Quantity	Baseline	Proposed (hybrid)
Physical-probe usage	100%	24.4% (measured)
Estimated probe life	6 months	~24.6 months (~4.11x)
Est. cost over 3 yrs	$360.0	$87.67
Estimated savings	—	$272.33 (75.6%)

ESTIMATE ONLY: measured 75.6% usage reduction + first-order wear∝usage assumption. Actual life depends on chemistry/temperature/handling, not directly measured here.